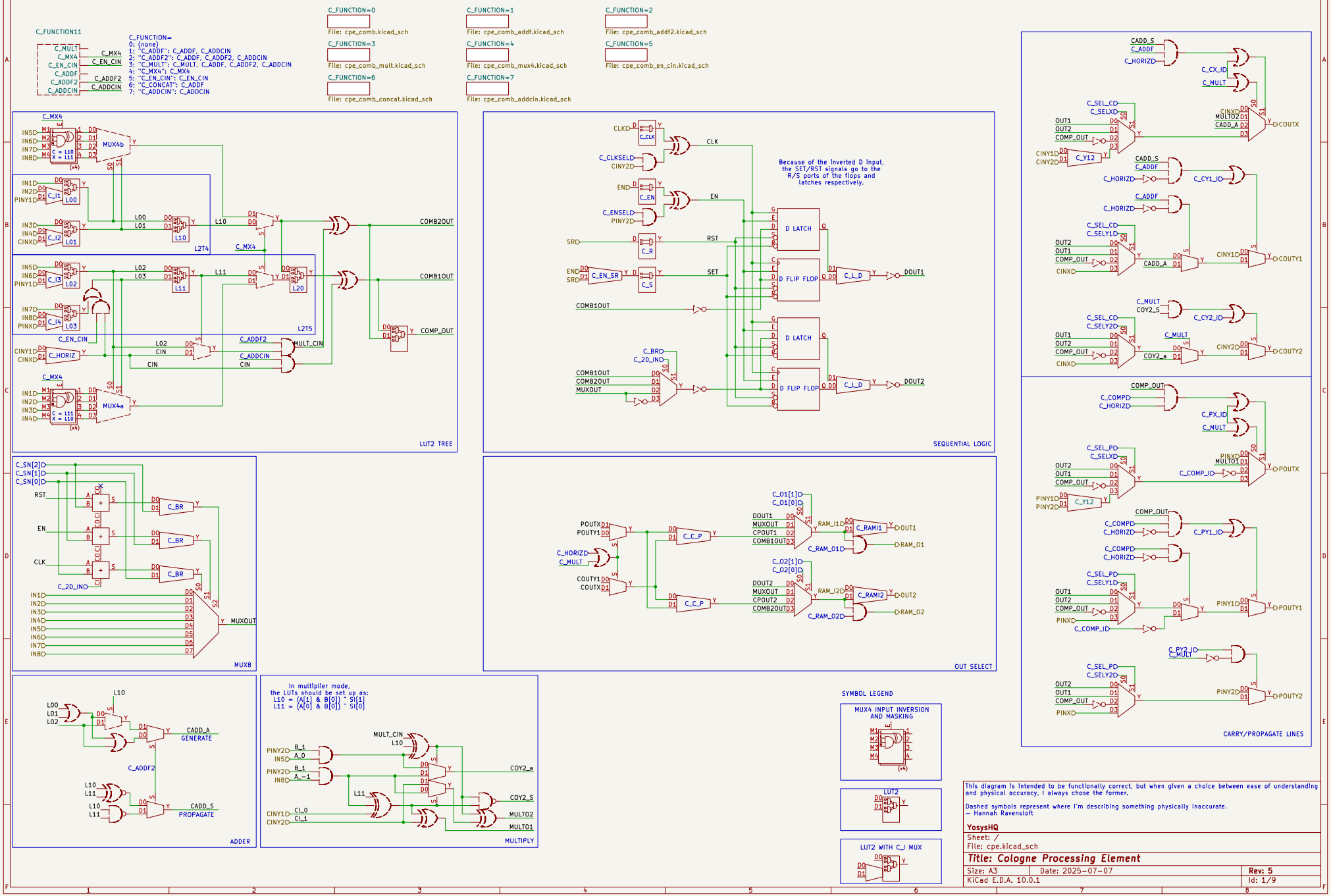

C_FUNCTION11
C_MULT
C_MX4
C_EN_CIN
C_ADDF
C_ADDF2
C_ADDCIN
C_MX4
C_EN_CIN
C_ADDF2
C_ADDCIN
C_FUNCTION=
0: (none)
1: "C_ADDF": C_ADDF, C_ADDCIN
2: "C_ADDF2": C_ADDF, C_ADDF2, C_ADDCIN
3: "C_MULT": C_MULT, C_ADDF, C_ADDF2, C_ADDCIN
4: "C_MX4": C_MX4
5: "C_EN_CIN": C_EN_CIN
6: "C_CONCAT": C_ADDF
7: "C_ADDCIN": C_ADDCIN
C_FUNCTION=0
File: cpe_comb.kicad_sch
C_FUNCTION=1
File: cpe_comb_addf.kicad_sch
C_FUNCTION=2
File: cpe_comb_addf2.kicad_sch
C_FUNCTION=3
File: cpe_comb_mult.kicad_sch
C_FUNCTION=4
File: cpe_comb_mux4.kicad_sch
C_FUNCTION=5
File: cpe_comb_en_cin.kicad_sch
C_FUNCTION=6
File: cpe_comb_concat.kicad_sch
C_FUNCTION=7
File: cpe_comb_addcin.kicad_sch
LUT2 TREE
MUX4b
MUX4a
L2T4
L2T5
COMB2OUT
COMB1OUT
COMP_OUT
MULT_CIN
SEQUENTIAL LOGIC
Because of the inverted D input, the SET/RST signals go to the R/S ports of the flops and latches respectively.
D LATCH
D FLIP FLOP
DOUT1
DOUT2
OUT SELECT
RAM_I1
RAM_I2
RAM_O1
RAM_O2
CARRY/PROPAGATE LINES
MUX8
MUXOUT
ADDER
CADD_A GENERATE
CADD_S PROPAGATE
MULTIPLY
In multiplier mode, the LUTs should be set up as:
L10 = (A[1] & B[0]) ^ SI[1]
L11 = (A[0] & B[0]) ^ SI[0]
COY2_a
COY2_S
MULTO2
MULTO1
SYMBOL LEGEND
MUX4 INPUT INVERSION AND MASKING
LUT2
LUT2 WITH C_I MUX
This diagram is intended to be functionally correct, but when given a choice between ease of understanding and physical accuracy, I always chose the former.
Dashed symbols represent where I'm describing something physically inaccurate.
– Hannah Ravensloft
YosysHQ
Sheet: /
File: cpe.kicad_sch
Title: Cologne Processing Element
Size: A3
Date: 2025-07-07
Rev: 5
KiCad E.D.A. 10.0.1
Id: 1/9